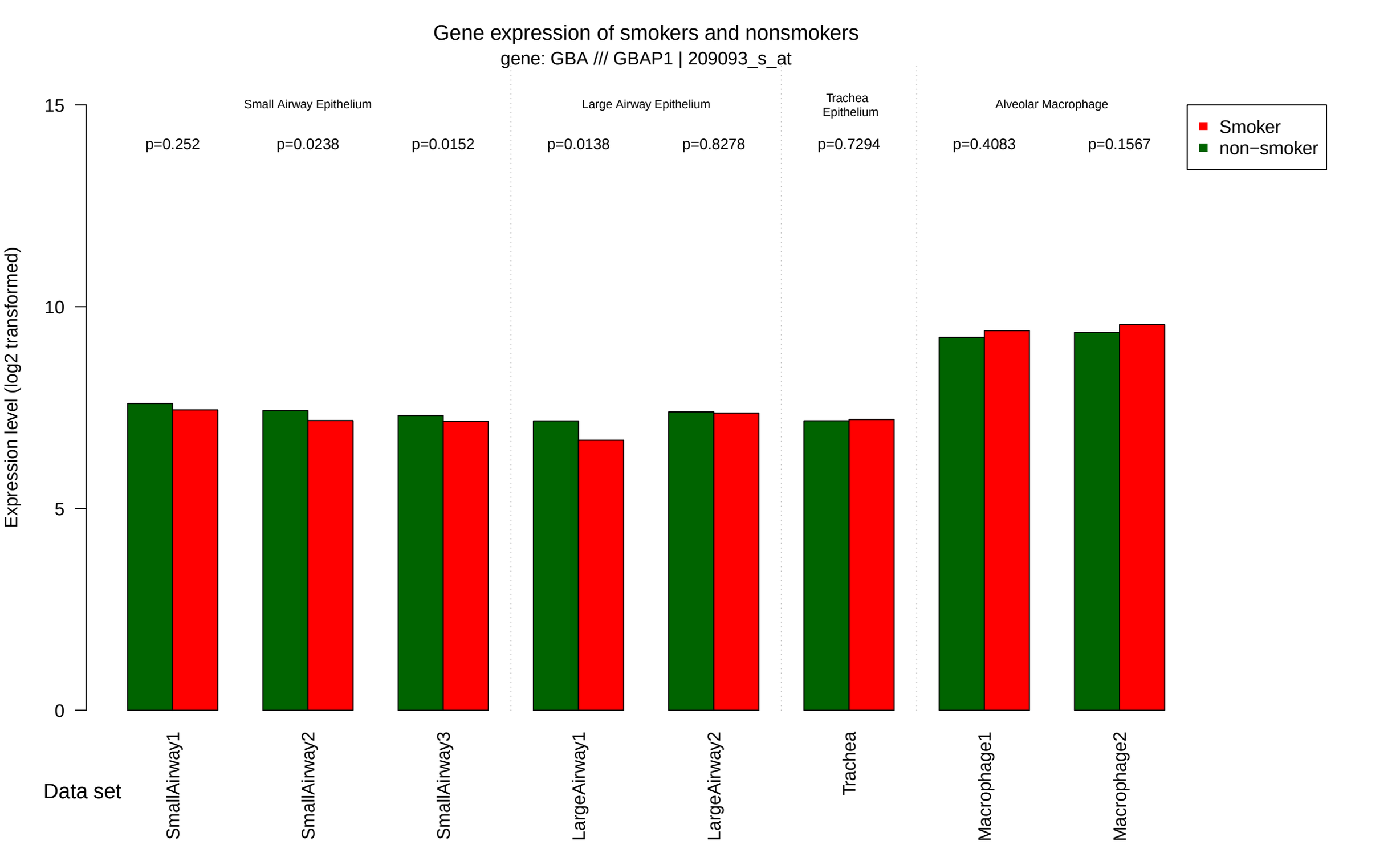
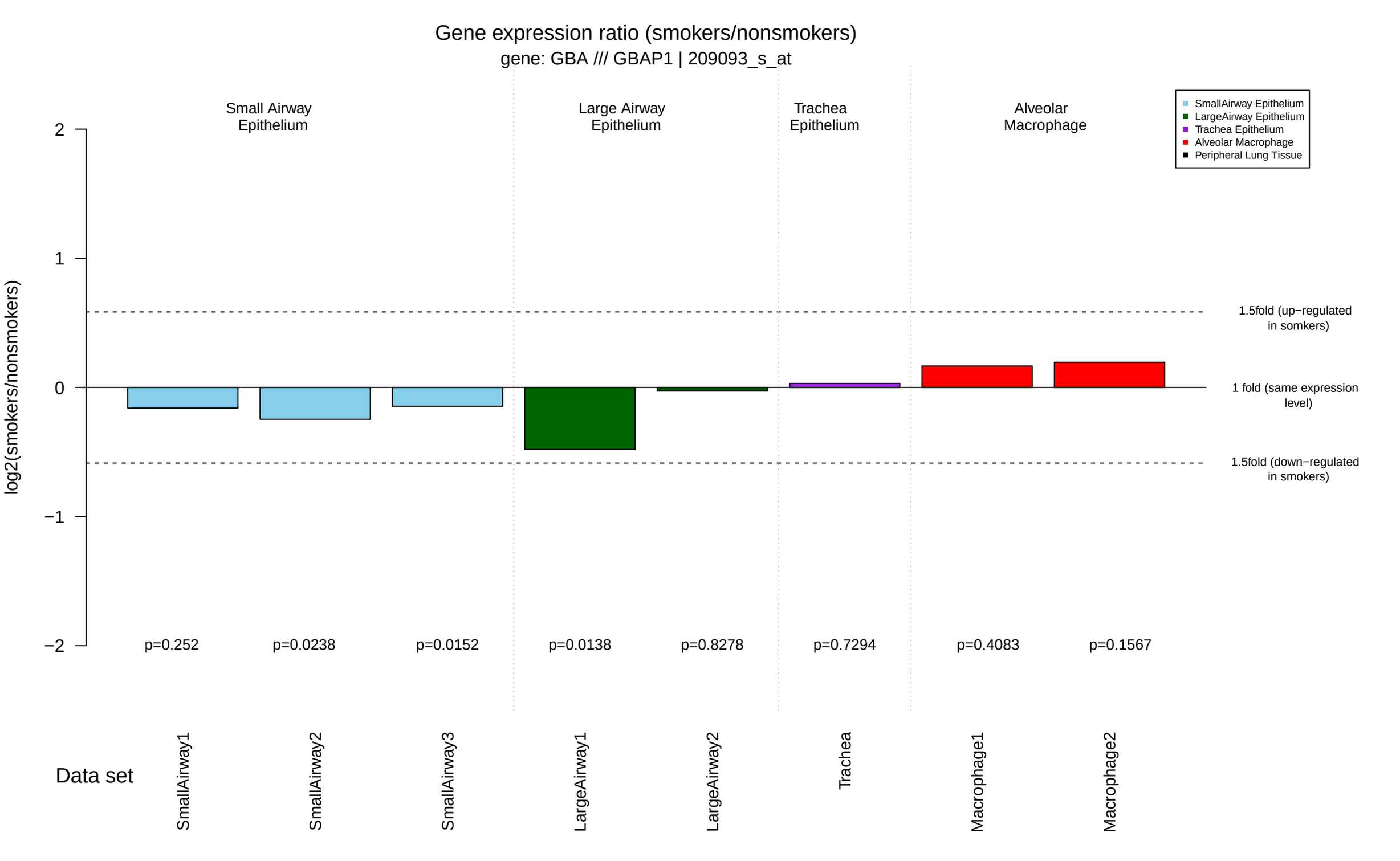
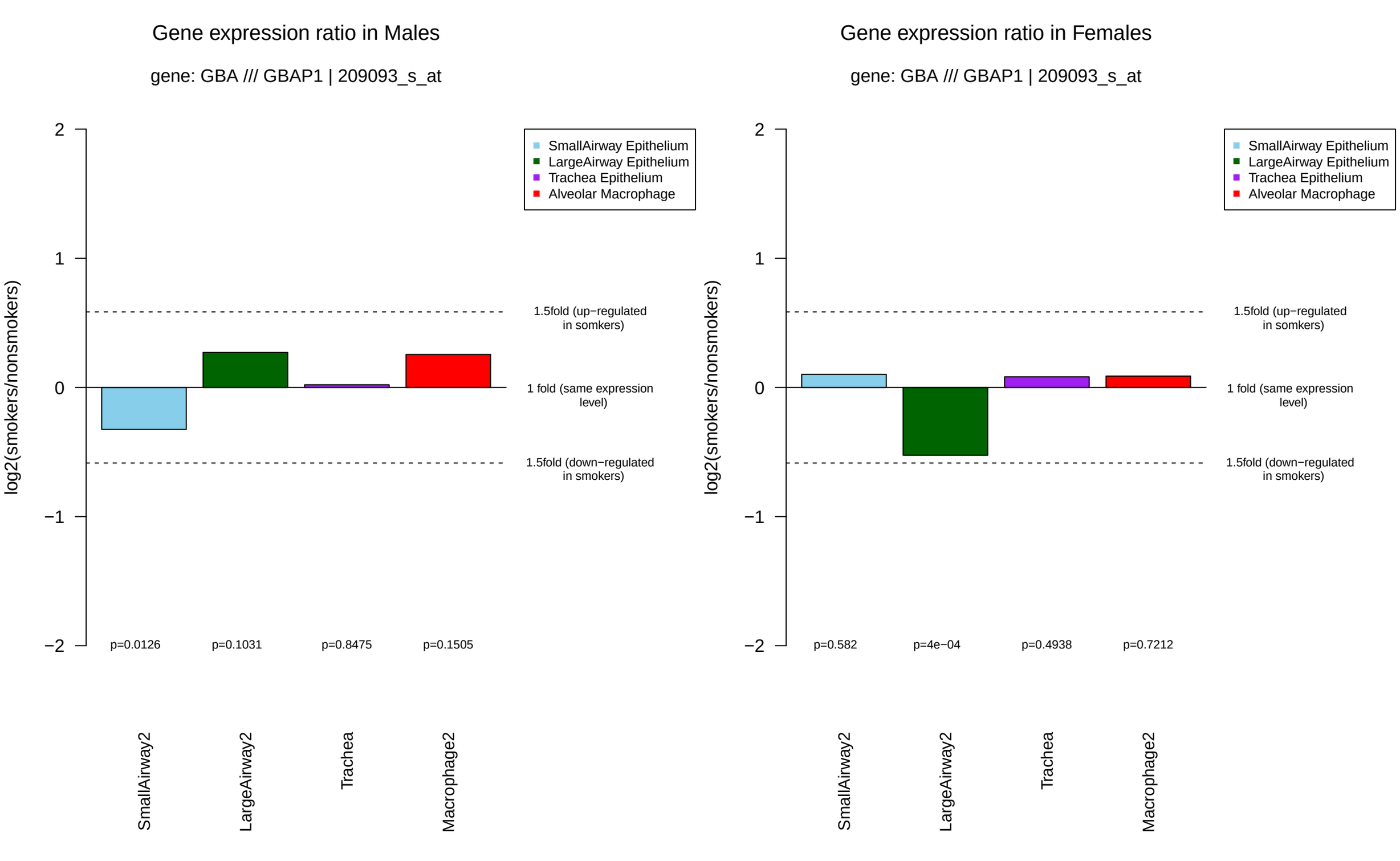
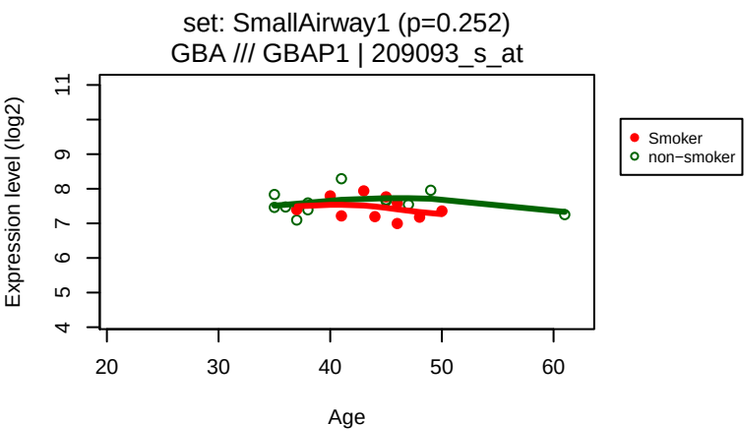
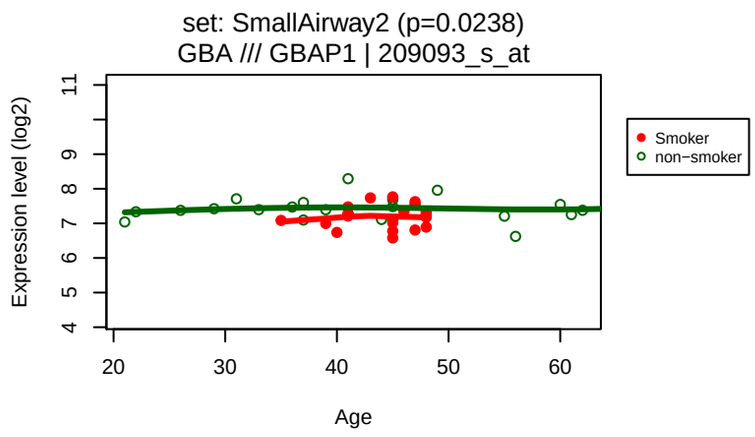
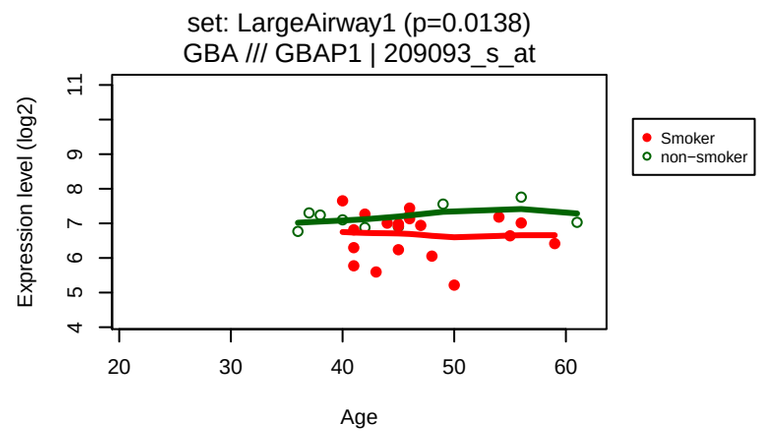

Gene expression of smokers and nonsmokers
gene: GBA /// GBAP1 | 209093_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.252
p=0.0238
p=0.0152
p=0.0138
p=0.8278
p=0.7294
p=0.4083
p=0.1567
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

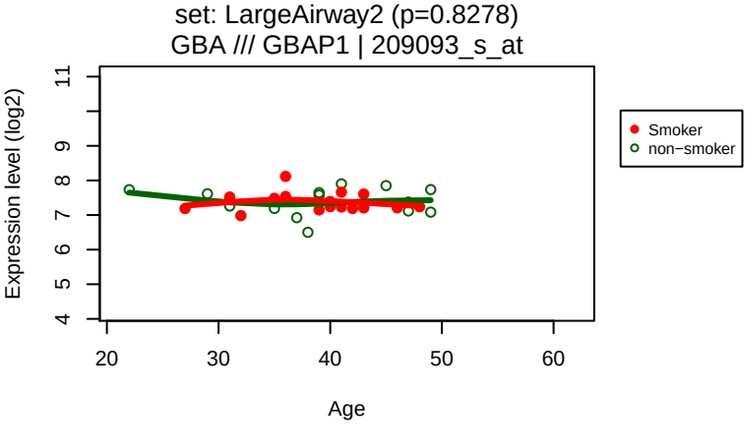

Gene expression ratio (smokers/nonsmokers)
gene: GBA /// GBAP1 | 209093_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.252
p=0.0238
p=0.0152
p=0.0138
p=0.8278
p=0.7294
p=0.4083
p=0.1567
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

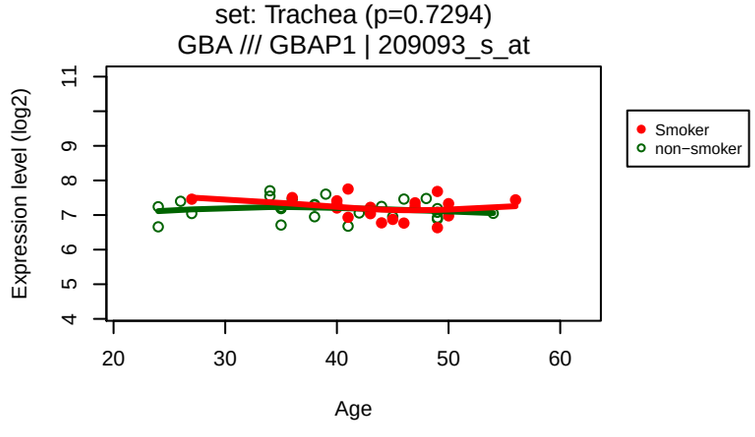

Gene expression ratio in Males
gene: GBA /// GBAP1 | 209093_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0126
p=0.1031
p=0.8475
p=0.1505
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GBA /// GBAP1 | 209093_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.582
p=4e−04
p=0.4938
p=0.7212
SmallAirway2
LargeAirway2
Trachea
Macrophage2

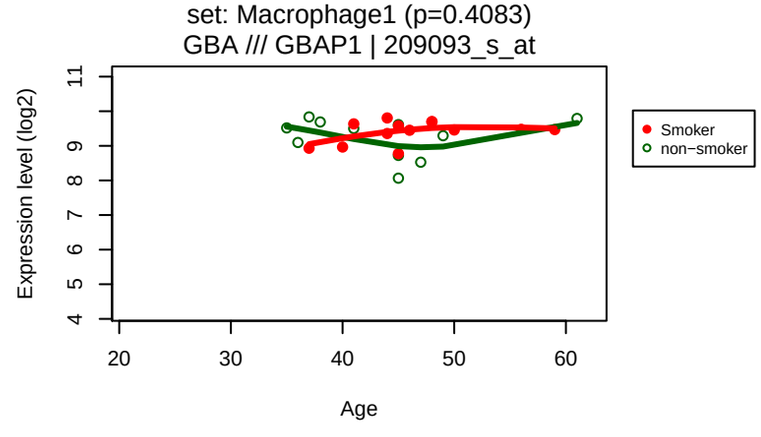

set: SmallAirway1 (p=0.252)
GBA /// GBAP1 | 209093_s_at
Expression level (log2)
Age
Smoker
non−smoker

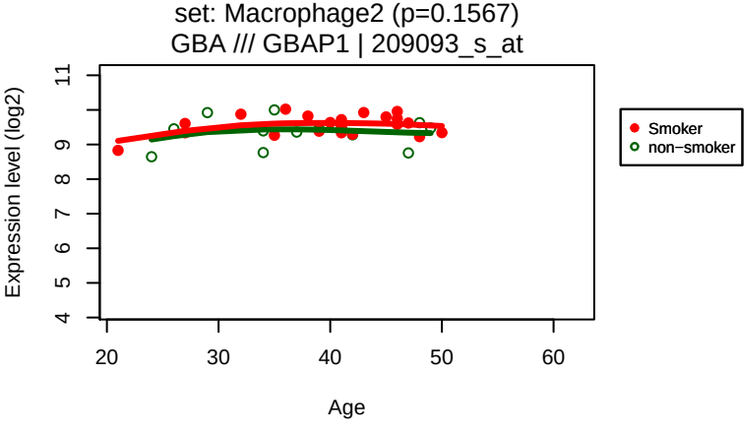

set: SmallAirway2 (p=0.0238)
GBA /// GBAP1 | 209093_s_at
Expression level (log2)
Age
Smoker
non−smoker

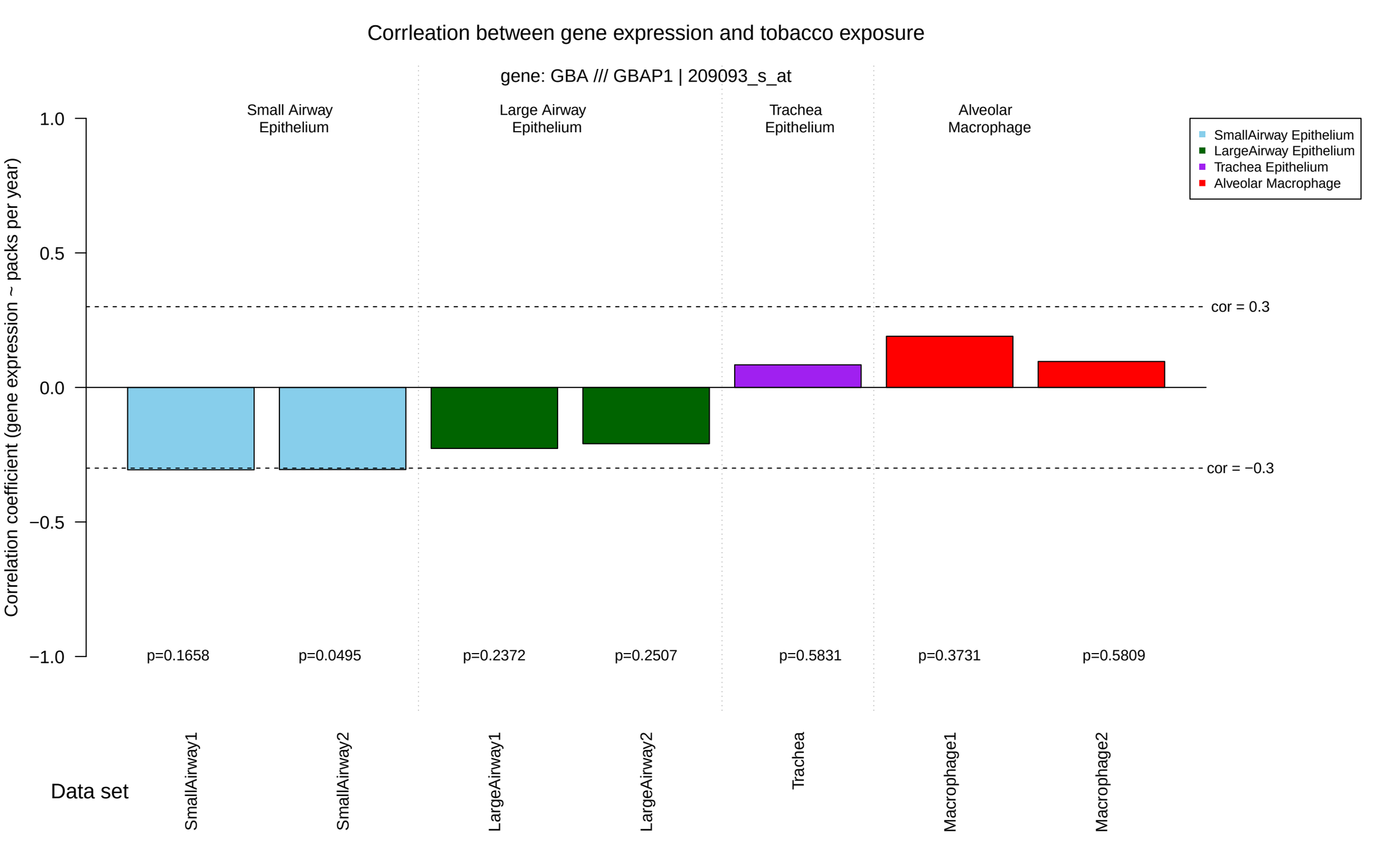

set: LargeAirway1 (p=0.0138)
GBA /// GBAP1 | 209093_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8278)
GBA /// GBAP1 | 209093_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7294)
GBA /// GBAP1 | 209093_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4083)
GBA /// GBAP1 | 209093_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1567)
GBA /// GBAP1 | 209093_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GBA /// GBAP1 | 209093_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1658
p=0.0495
p=0.2372
p=0.2507
p=0.5831
p=0.3731
p=0.5809
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2